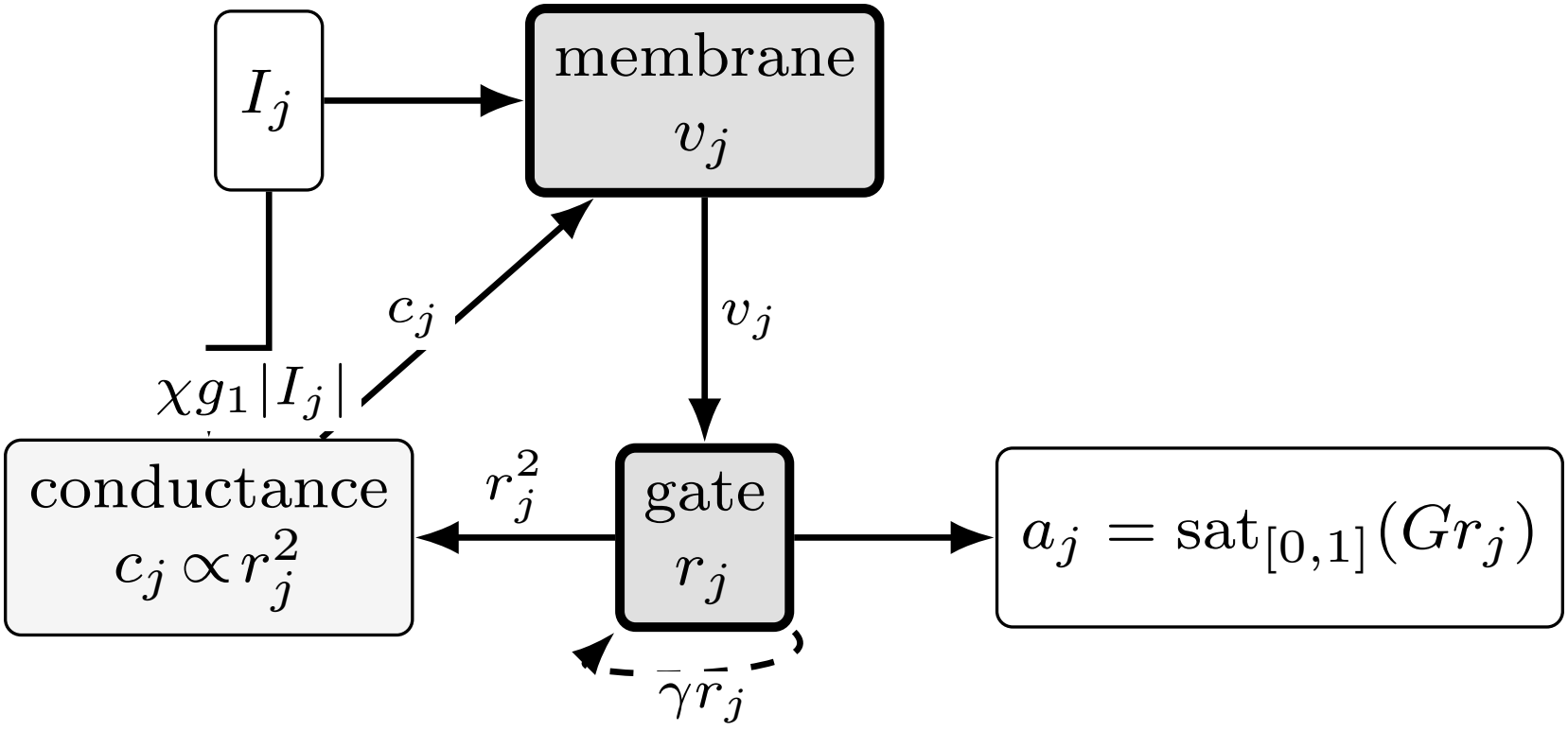
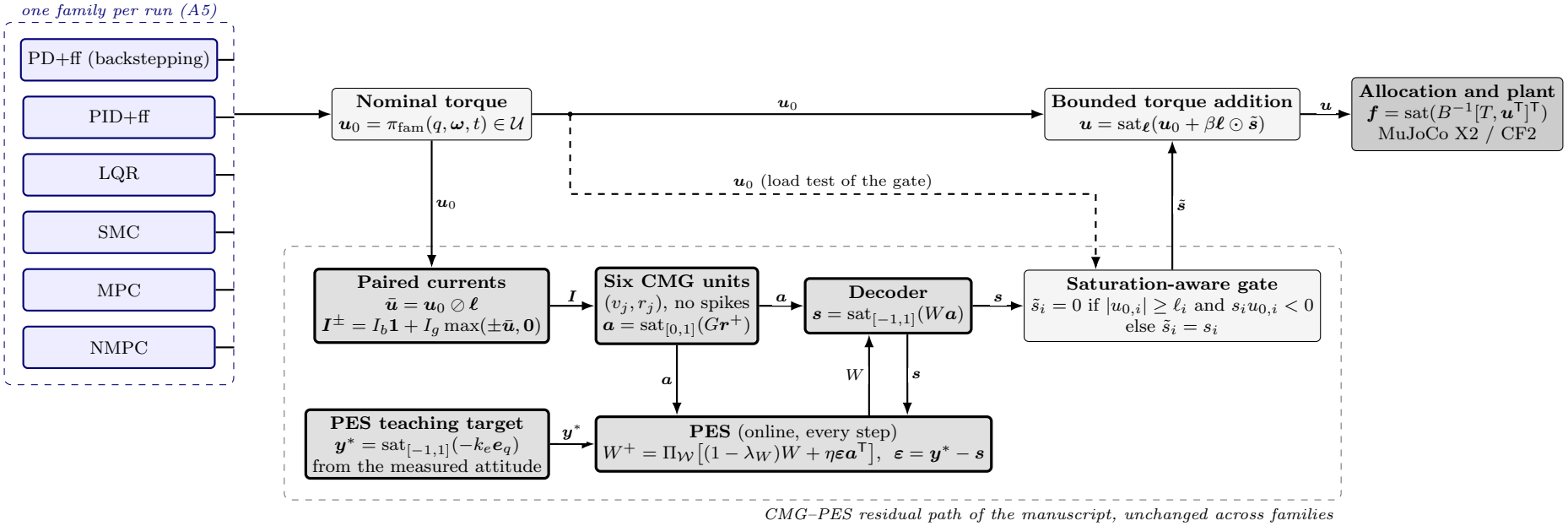
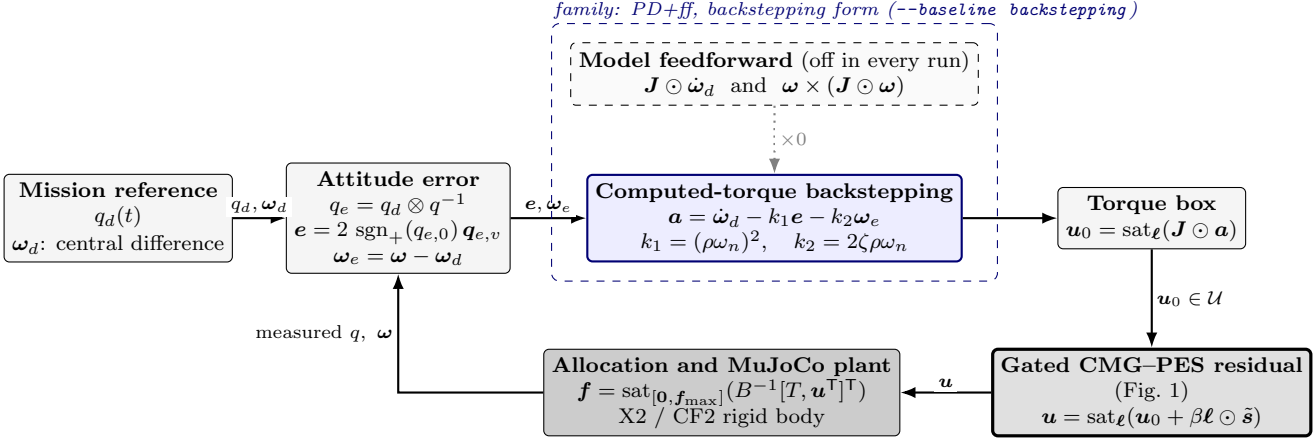
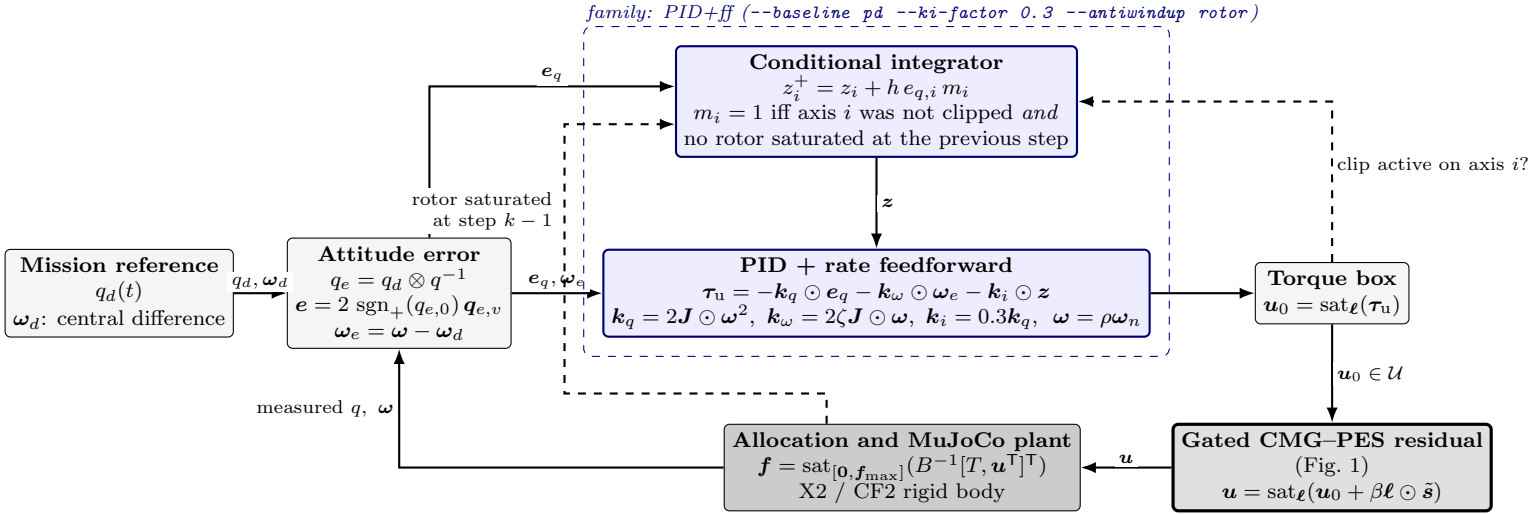
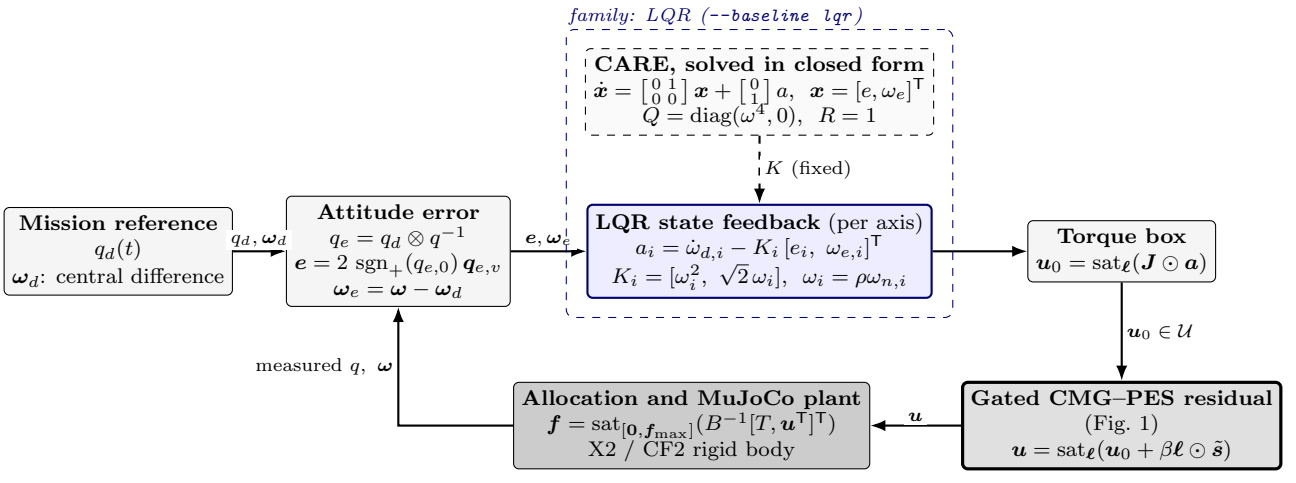
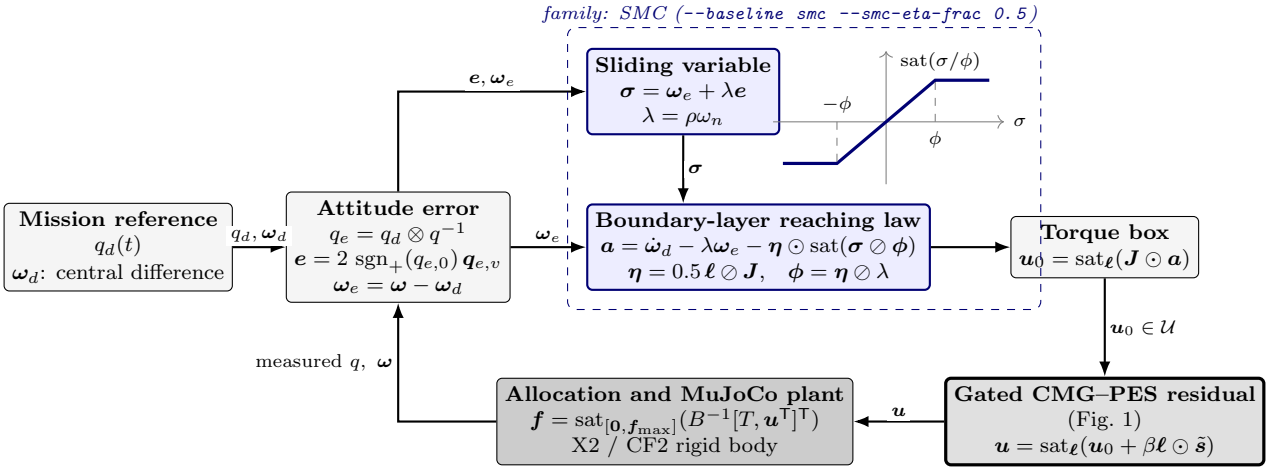

I_j
membrane
v_j
c_j
v_j
$\chi g_1 |I_j|$
conductance
$c_j \propto r_j^2$
r_j^2
gate
r_j
$a_j = \mathrm{sat}_{[0,1]}(Gr_j)$
$\bar{\gamma}\bar{r}_j$

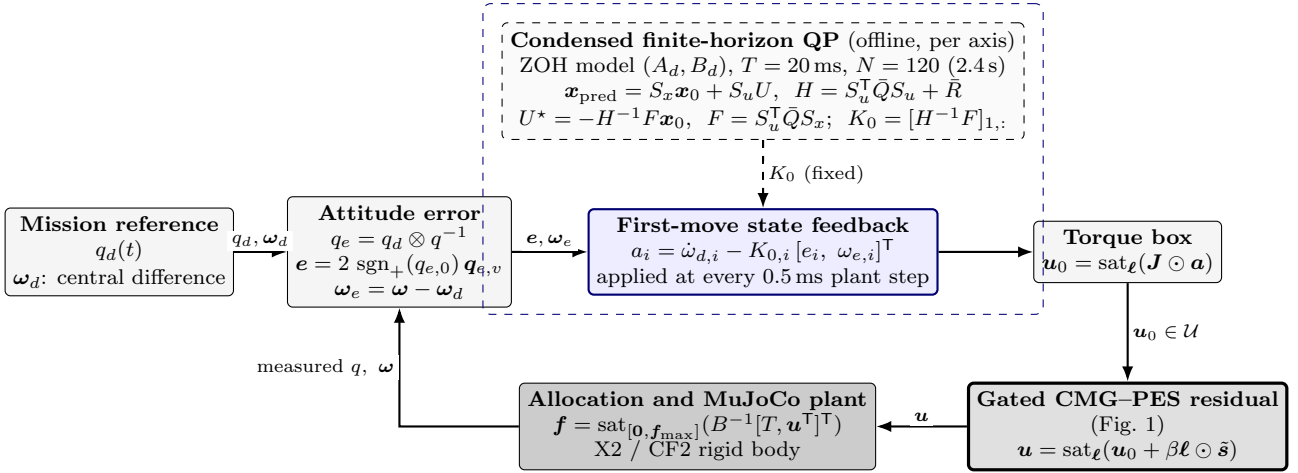

one family per run (A5)
PD+ff (backstepping)
PID+ff
LQR
SMC
MPC
NMPC
Nominal torque
$\boldsymbol{u}_0 = \pi_{\mathrm{fam}}(q, \boldsymbol{\omega}, t) \in \mathcal{U}$
$\boldsymbol{u}_0$
$\boldsymbol{u}_0$ (load test of the gate)
Bounded torque addition
$\boldsymbol{u} = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_0 + \beta \boldsymbol{\ell} \odot \tilde{\boldsymbol{s}})$
$\boldsymbol{u}$
Allocation and plant
$\boldsymbol{f} = \mathrm{sat}(B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}})$
MuJoCo X2 / CF2
$\tilde{\boldsymbol{s}}$
Paired currents
$\bar{\boldsymbol{u}} = \boldsymbol{u}_0 \oslash \boldsymbol{\ell}$
$\boldsymbol{I}^{\pm} = I_b \mathbf{1} + I_g \max(\pm\bar{\boldsymbol{u}}, \mathbf{0})$
$\boldsymbol{I}$
Six CMG units
(v_j, r_j), no spikes
$\boldsymbol{a} = \mathrm{sat}_{[0,1]}(G\boldsymbol{r}^{+})$
$\boldsymbol{a}$
Decoder
$\boldsymbol{s} = \mathrm{sat}_{[-1,1]}(W\boldsymbol{a})$
$\boldsymbol{s}$
Saturation-aware gate
$\tilde{s}_i = 0$ if $|u_{0,i}| \geq \ell_i$ and $s_i u_{0,i} < 0$
else $\tilde{s}_i = s_i$
$\boldsymbol{a}$
W
$\boldsymbol{s}$
PES teaching target
$\boldsymbol{y}^{*} = \mathrm{sat}_{[-1,1]}(-k_e \boldsymbol{e}_q)$
from the measured attitude
$\boldsymbol{y}^{*}$
PES (online, every step)
$W^{+} = \Pi_{\mathcal{W}}\left[(1 - \lambda_W)W + \eta \boldsymbol{\varepsilon} \boldsymbol{a}^{\mathsf{T}}\right], \ \boldsymbol{\varepsilon} = \boldsymbol{y}^{*} - \boldsymbol{s}$
CMG–PES residual path of the manuscript, unchanged across families

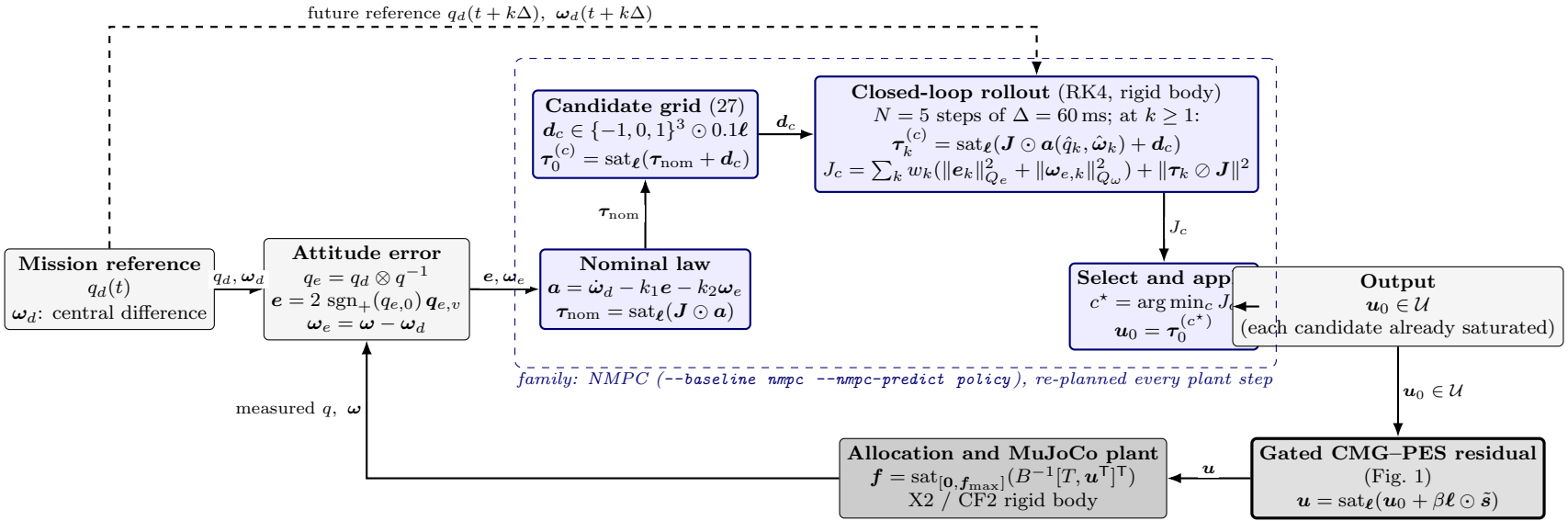

family: PD+ff, backstepping form (--baseline backstepping)
Model feedforward (off in every run)
$\boldsymbol{J} \odot \dot{\boldsymbol{\omega}}_d$ and $\boldsymbol{\omega} \times (\boldsymbol{J} \odot \boldsymbol{\omega})$
$\times 0$
Mission reference
$q_d(t)$
$\boldsymbol{\omega}_d$: central difference
$q_d, \boldsymbol{\omega}_d$
Attitude error
$q_e = q_d \otimes q^{-1}$
$\boldsymbol{e} = 2\ \mathrm{sgn}_+(q_{e,0})\, \boldsymbol{q}_{e,v}$
$\boldsymbol{\omega}_e = \boldsymbol{\omega} - \boldsymbol{\omega}_d$
$\boldsymbol{e}, \dot{\boldsymbol{\omega}}_e$
Computed-torque backstepping
$\boldsymbol{a} = \ddot{\boldsymbol{\omega}}_d - k_1 \boldsymbol{e} - k_2 \boldsymbol{\omega}_e$
$k_1 = (\rho\omega_n)^2, \quad k_2 = 2\zeta\rho\omega_n$
Torque box
$\boldsymbol{u}_0 = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{J} \odot \boldsymbol{a})$
$\boldsymbol{u}_0 \in \mathcal{U}$
Gated CMG–PES residual
(Fig. 1)
$\boldsymbol{u} = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_0 + \beta\boldsymbol{\ell} \odot \tilde{\boldsymbol{s}})$
$\boldsymbol{u}$
Allocation and MuJoCo plant
$\boldsymbol{f} = \mathrm{sat}_{[\mathbf{0}, \boldsymbol{f}_{\max}]}(B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}})$
X2 / CF2 rigid body
measured q, $\boldsymbol{\omega}$


family: PID+ff (--baseline pd --ki-factor 0.3 --antiwindup rotor)
Conditional integrator
$z_i^+ = z_i + h\, e_{q,i}\, m_i$
$m_i = 1$ iff axis i was not clipped and no rotor saturated at the previous step
e_q
z
clip active on axis i?
rotor saturated at step $k-1$
Mission reference
$q_d(t)$
ω_d: central difference
q_d, ω_d
Attitude error
$q_e = q_d \otimes q^{-1}$
$e = 2\ \mathrm{sgn}_+(q_{e,0})\, q_{e,v}$
$\omega_e = \omega - \omega_d$
e_q, ω_e
PID + rate feedforward
$\tau_u = -k_q \odot e_q - k_\omega \odot \omega_e - k_i \odot z$
$k_q = 2J \odot \omega^2,\ k_\omega = 2\zeta J \odot \omega,\ k_i = 0.3 k_q,\ \omega = \rho \omega_n$
Torque box
$u_0 = \mathrm{sat}_\ell(\tau_u)$
$u_0 \in \mathcal{U}$
measured q, ω
Allocation and MuJoCo plant
$f = \mathrm{sat}_{[0, f_{max}]}(B^{-1}[T, u^T]^T)$
X2 / CF2 rigid body
u
Gated CMG–PES residual
(Fig. 1)
$u = \mathrm{sat}_\ell(u_0 + \beta \ell \odot \tilde{s})$


family: LQR (--baseline lqr)
CARE, solved in closed form
$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} a$, $x = [e, \omega_e]^\mathsf{T}$
$Q = \mathrm{diag}(\omega^4, 0)$, $R = 1$
K (fixed)
Mission reference
$q_d(t)$
ω_d: central difference
q_d, ω_d
Attitude error
$q_e = q_d \otimes q^{-1}$
$e = 2\ \mathrm{sgn}_+(q_{e,0})\, q_{e,v}$
$\omega_e = \omega - \omega_d$
e, ω_e
LQR state feedback (per axis)
$a_i = \dot{\omega}_{d,i} - K_i\,[e_i,\ \omega_{e,i}]^\mathsf{T}$
$K_i = [\omega_i^2,\ \sqrt{2}\,\omega_i]$, $\omega_i = \rho\omega_{n,i}$
Torque box
$u_0 = \mathrm{sat}_{\ell}(J \odot a)$
$u_0 \in \mathcal{U}$
Gated CMG–PES residual
(Fig. 1)
$u = \mathrm{sat}_{\ell}(u_0 + \beta\ell \odot \tilde{s})$
u
Allocation and MuJoCo plant
$f = \mathrm{sat}_{[0, f_{\max}]}(B^{-1}[T, u^\mathsf{T}]^\mathsf{T})$
X2 / CF2 rigid body
measured q, ω


family: SMC (--baseline smc --smc-eta-frac 0.5)
Mission reference
$q_d(t)$
$\boldsymbol{\omega}_d$: central difference
$q_d, \boldsymbol{\omega}_d$
Attitude error
$q_e = q_d \otimes q^{-1}$
$\boldsymbol{e} = 2\ \mathrm{sgn}_+(q_{e,0})\, \boldsymbol{q}_{e,v}$
$\boldsymbol{\omega}_e = \boldsymbol{\omega} - \boldsymbol{\omega}_d$
$\boldsymbol{e}, \boldsymbol{\omega}_e$
$\boldsymbol{\omega}_e$
Sliding variable
$\boldsymbol{\sigma} = \boldsymbol{\omega}_e + \lambda \boldsymbol{e}$
$\lambda = \rho \omega_n$
$\boldsymbol{\sigma}$
$\mathrm{sat}(\sigma/\phi)$
$-\phi$
ϕ
σ
Boundary-layer reaching law
$\boldsymbol{a} = \dot{\boldsymbol{\omega}}_d - \lambda \boldsymbol{\omega}_e - \boldsymbol{\eta} \odot \mathrm{sat}(\boldsymbol{\sigma} \oslash \boldsymbol{\phi})$
$\boldsymbol{\eta} = 0.5\, \boldsymbol{\ell} \oslash \boldsymbol{J}, \quad \boldsymbol{\phi} = \boldsymbol{\eta} \oslash \lambda$
Torque box
$\boldsymbol{u}_0 = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{J} \odot \boldsymbol{a})$
$\boldsymbol{u}_0 \in \mathcal{U}$
Gated CMG–PES residual
(Fig. 1)
$\boldsymbol{u} = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_0 + \beta \boldsymbol{\ell} \odot \tilde{\boldsymbol{s}})$
$\boldsymbol{u}$
Allocation and MuJoCo plant
$\boldsymbol{f} = \mathrm{sat}_{[\boldsymbol{0}, \boldsymbol{f}_{\max}]}(B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}})$
X2 / CF2 rigid body
measured q, $\boldsymbol{\omega}$


family: MPC (--baseline mpc), receding horizon realized as its first-move gain
Condensed finite-horizon QP (offline, per axis)
ZOH model (A_d, B_d), $T = 20\,\mathrm{ms}$, $N = 120$ (2.4 s)
$\boldsymbol{x}_{\mathrm{pred}} = S_x \boldsymbol{x}_0 + S_u U,\ \ H = S_u^{\mathsf{T}} \bar{Q} S_u + \bar{R}$
$U^\star = -H^{-1} F \boldsymbol{x}_0,\ \ F = S_u^{\mathsf{T}} \bar{Q} S_x;\ \ K_0 = [H^{-1} F]_{1,:}$
K_0 (fixed)
Mission reference
$q_d(t)$
$\boldsymbol{\omega}_d$: central difference
$q_d, \boldsymbol{\omega}_d$
Attitude error
$q_e = q_d \otimes q^{-1}$
$\boldsymbol{e} = 2\ \mathrm{sgn}_+(q_{e,0})\ \boldsymbol{q}_{e,v}$
$\boldsymbol{\omega}_e = \boldsymbol{\omega} - \boldsymbol{\omega}_d$
$\boldsymbol{e}, \boldsymbol{\omega}_e$
First-move state feedback
$a_i = \dot{\omega}_{d,i} - K_{0,i}\,[e_i,\ \omega_{e,i}]^{\mathsf{T}}$
applied at every 0.5 ms plant step
Torque box
$\boldsymbol{u}_0 = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{J} \odot \boldsymbol{a})$
$\boldsymbol{u}_0 \in \mathcal{U}$
Gated CMG–PES residual
(Fig. 1)
$\boldsymbol{u} = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_0 + \beta \boldsymbol{\ell} \odot \tilde{\boldsymbol{s}})$
$\boldsymbol{u}$
Allocation and MuJoCo plant
$\boldsymbol{f} = \mathrm{sat}_{[\boldsymbol{0}, \boldsymbol{f}_{\max}]}(B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}})$
X2 / CF2 rigid body
measured $q,\ \boldsymbol{\omega}$


future reference $q_d(t+k\Delta)$, $\boldsymbol{\omega}_d(t+k\Delta)$
Mission reference
$q_d(t)$
$\boldsymbol{\omega}_d$: central difference
$q_d, \boldsymbol{\omega}_d$
Attitude error
$q_e = q_d \otimes q^{-1}$
$\boldsymbol{e} = 2\ \mathrm{sgn}_+(q_{e,0})\, \boldsymbol{q}_{e,v}$
$\boldsymbol{\omega}_e = \boldsymbol{\omega} - \boldsymbol{\omega}_d$
$\boldsymbol{e}, \boldsymbol{\omega}_e$
Nominal law
$\boldsymbol{a} = \ddot{\boldsymbol{\omega}}_d - k_1\boldsymbol{e} - k_2\boldsymbol{\omega}_e$
$\boldsymbol{\tau}_{\mathrm{nom}} = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{J} \odot \boldsymbol{a})$
$\boldsymbol{\tau}_{\mathrm{nom}}$
Candidate grid (27)
$\boldsymbol{d}_c \in \{-1, 0, 1\}^3 \odot 0.1\boldsymbol{\ell}$
$\boldsymbol{\tau}_0^{(c)} = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{\tau}_{\mathrm{nom}} + \boldsymbol{d}_c)$
$\boldsymbol{d}_c$
Closed-loop rollout (RK4, rigid body)
$N = 5$ steps of $\Delta = 60$ ms; at $k \geq 1$:
$\boldsymbol{\tau}_k^{(c)} = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{J} \odot \boldsymbol{a}(\hat{q}_k, \hat{\boldsymbol{\omega}}_k) + \boldsymbol{d}_c)$
$J_c = \sum_k w_k(\|\boldsymbol{e}_k\|^2_{Q_e} + \|\boldsymbol{\omega}_{e,k}\|^2_{Q_\omega}) + \|\boldsymbol{\tau}_k \oslash \boldsymbol{J}\|^2$
J_c
Select and app
$c^\star = \arg\min_c J_c$
$\boldsymbol{u}_0 = \boldsymbol{\tau}_0^{(c^\star)}$
Output
$\boldsymbol{u}_0 \in \mathcal{U}$
(each candidate already saturated)
family: NMPC (--baseline nmpc --nmpc-predict policy), re-planned every plant step
$\boldsymbol{u}_0 \in \mathcal{U}$
Gated CMG–PES residual
(Fig. 1)
$\boldsymbol{u} = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_0 + \beta\boldsymbol{\ell} \odot \hat{\boldsymbol{s}})$
$\boldsymbol{u}$
Allocation and MuJoCo plant
$\boldsymbol{f} = \mathrm{sat}_{[\boldsymbol{0}, \boldsymbol{f}_{\max}]}(B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}})$
X2 / CF2 rigid body
measured q, $\boldsymbol{\omega}$